R&D | RetailPulse

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Hypothesis and Insight Playbook

> **In simple words**
>
> This document teaches how to turn SQL numbers into insights and actions. It gives the sentence formula, the insight card, a list of hypotheses to test, and a log to fill in.

1. What is an insight?

An **insight** is a finding that explains something and leads to action. It answers **What? Compared with what? Why? So what?**

Not an insight	An insight
WAPE for the family is [X]%.	WAPE for the family is [X] points higher than overall, and most of it comes from the week after holidays. Do not copy holiday sales into next week's plan.
December has more sales.	In December the baseline is [too low / too high] by [X]%, so plan buffer stock for [family].

Anything in [brackets] is a placeholder. Replace it with your real result.

2. Insight sentence formula

> **Use this every time**
>
> **[Where / when]** has **[metric] of [X]**, which is **[higher / lower] than [comparison]**. This is **likely because [reason]**. We recommend **[action]**, owned by **[team]**.

3. Insight card (one per finding)

Field	Write here
ID and title	I-01: short headline
Question answered	Which business question from Doc 01 (A to E)
Evidence	Key numbers and the chart / query that shows it
Comparison	Compared with overall, another group, or another period
Likely reason	Hypothesis result: confirmed, likely, or needs more data
Recommendation	Specific action
Owner and priority	Team and High / Medium / Low
Expected effect	In units or WAPE points, not money
Limit	What this does not prove

4. Hypotheses to test

These are starting guesses, **not facts**. Test each with the SQL question shown. A rejected hypothesis is still a useful finding.

ID	Hypothesis	Why we think so	Test	If true, possible action
H1	Error is higher on holidays	Demand jumps and last week is normal	Q7	Holiday-specific forecast or buffer
H2	One week after a holiday the baseline is too high	It copies the holiday spike	Q7	Do not use holiday weeks as the base
H3	December is the hardest month	Christmas shopping	Q6	Plan December separately
H4	Low-volume families have the highest WAPE	Small numbers are noisy	Q2	Track them, do not over-fix
H5	A few families cause most of the error	Big families dominate units	Q3	Fix the top families first
H6	Promotions raise error and cause shortfall	Sales jump above last week	Q8	Feed promotion plan into the forecast
H7	Paydays (15th and last day) raise sales above baseline	Wages paid on those days	Q9	Payday calendar adjustment
H8	Weekends have more error	Shopping patterns differ	Q10	Weekday adjustment
H9	Smaller cities and some store types have higher WAPE	Lower volume, more noise	Q5	Different rules by store type
H10	A 4-week average beats last week for volatile families	Averaging reduces noise	Q11	Use the better baseline by family

5. Confidence labels

Label	Meaning
Confirmed by data	Clear pattern, large size, holds across groups
Likely	Pattern is clear, but the cause is our guess
Needs more data	Small groups or mixed results

Correlation is not causation. A pattern shows that two things move together. It does not prove one causes the other.

6. Types of recommendation

Type	Use when	Example form
Adjust the method	A simple baseline fails in a clear way	Use a 4-week average for [family]
Add buffer stock	Shortfall units are high	Add [X]% buffer for [family] in [month]
Reduce orders	Excess units are high	Lower the order for [family] in [region]
Plan events	Holidays, paydays, promotions break the pattern	Add an event calendar to planning
Monitor	Evidence is weak or volume is tiny	Track weekly, review next quarter
Fix data	Data gaps cause error	Record store opening dates

Rank actions by **impact** (share of error or units) and **effort** (easy or hard). Do high impact and low effort first.

7. Insight log (fill in as you analyze)

ID	Question / SQL	Finding with numbers	Reason	Action	Confidence
I-01					
I-02					
I-03					
I-04					
I-05					
I-06					
I-07					
I-08					

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